



Project title

by

Your Name

A dissertation submitted in fulfilment of the requirements of the degree of

Master of Commerce

in

Information Technology Management

at the

College of Business and Economics

UNIVERSITY OF JOHANNESBURG

Supervisor: Dr. Siyabonga Mhlongo

2026

Abstract

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Keywords: Keyword one, Keyword two, Keyword 3.

Declaration

I, Your Name, certify that the dissertation submitted by me for the Master of Commerce in Information Technology Management degree at the University of Johannesburg is my independent work and has not been submitted by me for a degree at another university.

YOUR NAME

(No signature required)

Dedication

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Acknowledgements

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Contents

Abstract	i
Declaration	ii
Dedication	iii
Acknowledgements	iv
List of Acronyms	vii
List of Figures	viii
1 Introduction	1
1.1 Introduction	1
1.2 Background and context	1
1.3 Rationale of the study	3
1.4 Problem statement	3
1.5 Significance of the study	3
1.6 Overview of the theoretical/conceptual framework	3
1.7 Research aim and objectives	3
1.8 Research questions	3
1.9 Scope and delimitation	3
1.10 Methodology overview	3
1.11 Ethical considerations	3
1.12 Dissertation structure	3
1.13 Chapter summary	3

2 Literature Review	4
3 Research Methodology	5
4 Results	6
5 Discussion	7
6 Conclusion	8
References	9
A Ethical Clearance Certificate	11
B Appendix Chapter	14
B.1 Appendix section	14
C Another Appendix Chapter	15

List of Figures

Chapter 1

Introduction

1.1 Introduction

This chapter introduces the study and also provides the background for examining the role of IT leadership in achieving ROI from AI coding assistants. It outlines the research problem being addressed and discusses the significance of the topic. Furthermore the chapter also presents the research questions and objections of this study. The final section of this chapter will provide an overview of the research methodology and theoretical framework used to guide the study.

1.2 Background and context

Artificial intelligence (AI) is one of the structural factors that has a significant impact on productivity and competitive advantages (Gillespie et al., 2025). As of 2023, generative AI systems have developed from being an experiment to reaching the phase of implementation within organizational processes (Gillespie et al., 2025). Recent reports from around the world show that organizations allocate more money to AI since they expect clear monetary outcomes, cost savings, and improved performance of organizational processes (Gillespie et al., 2025). Research on information systems and management has transcended from the stage of adoption to the understanding of added value meanings with the emphasis on differentiation of organizational value created from investments in AI systems as a function of organizational governance and management performance, not technology excellence (Hossain et al., 2025).

In software engineering, the use of generative AI tools are among the most noticeable forms of

AI-enabled augmentation (Peng et al., 2023). Providing developers with functionalities such as code suggestions and entire code functions in real time based on context, documentation, and debugging (Peng et al., 2023). Practical controlled experiments provide conclusive evidence that AI-assisted developers complete software development tasks faster and report increased productivity of about 55.8% compared with control groups (Peng et al., 2023). Collaboration between humans and AI systems is a paradigm shift from the traditional concept of automation, in which AI enhances developers' capabilities rather than replacing them with machines (Peng et al., 2023). Initial performance improvements demonstrated in controlled promising experiments may not necessarily guarantee long-term organizational performance benefits. Empirical studies report considerable variation in the productivity gains from AI-assisted software development tasks across software development teams and organizations.

As it has been long noted, organization culture and management style have been generally believed to affect the implementation success of new technologies. Within the digital sphere, values and practices will drive the culture of experimentation, learning, and data use in decision making as well as a positive attitude to tech adoption. New findings reveal that digital culture explains the linkage between human AI skills and innovation outcomes, and that culture readiness mediates the performance impact of AI Investments (An et al., 2024). Similarly, this applied also to employees' engagement in AI, ethical AI management or leadership impacts toward strategic alignment. Even though the significance of digital culture and leadership is being gradually recognized by most researchers, current research findings on AI Adoption is more fragmented (An et al., 2024). A significant number of recent works investigated AI adoption under the macro-organizational view or focused on the impacts for personal performance only instead of integrating digital culture, management and ROI in the software industry (Peng et al., 2023).

Generative AI (GenAI) has been increasingly adopted by african companies to address resource constraints and operational inefficiencies. Similarly, there is evidence that developing nations have tried extensively with AI and have positive views about AI's capabilities to boost productivity at work (Gillespie et al., 2025). Yet, literature to date highlights difficulties around digital skills pipelines, governance maturity and leadership capability among others in adopting GenAI technologies (Hughes et al., 2026). In this context, organisational culture and leadership remain particularly critical. If these factors are not supported by a clear strategic vision, trust and a

learning orientation, AI adoption is likely to be less about value creation and more about segmentation or symbolism. The digital transformation landscape in Africa is a valid environment for exploring the impact of socio-organizational factors on the achievement of returns from AI technologies.

1.3 Rationale of the study

1.4 Problem statement

1.5 Significance of the study

1.6 Overview of the theoretical/conceptual framework

1.7 Research aim and objectives

1.8 Research questions

1.9 Scope and delimitation

1.10 Methodology overview

1.11 Ethical considerations

1.12 Dissertation structure

1.13 Chapter summary

Chapter 2

Literature Review

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Chapter 3

Research Methodology

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Chapter 4

Results

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Chapter 5

Discussion

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Chapter 6

Conclusion

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

References

- Alenezi, M., & Akour, M. (2025). Ai-driven innovations in software engineering: A review of current practices and future directions. *Applied Sciences*, 15(3), 1344. <https://doi.org/10.3390/app15031344>
- An, M., Lin, J., & Luo, X. (2024). The impact of human ai skills on organizational innovation: The moderating role of digital organizational culture. *Journal of Business Research*, 182, 114786. <https://doi.org/10.1016/j.jbusres.2024.114786>
- Gillespie, N., Lockey, S., Ward, T., Macdade, A., & Hased, G. (2025). *Trust, attitudes and use of artificial intelligence: A global study 2025* (tech. rep.). The University of Melbourne. The University of Melbourne. <https://doi.org/10.26188/28822919.V1>
- Hossain, S., Fernando, M., & Akter, S. (2025). The influence of artificial intelligence-driven capabilities on responsible leadership: A future research agenda. *Journal of Management & Organization*, 31(5), 2360–2384. <https://doi.org/10.1017/jmo.2025.10010>
- Hughes, L., Davies, F., Li, K., Gunaratnege, S. M., Malik, T., & Dwivedi, Y. K. (2026). Beyond the hype: Organisational adoption of generative ai through the lens of the toe framework—a mixed methods perspective. *International Journal of Information Management*, 86, 102982. <https://doi.org/10.1016/j.ijinfomgt.2025.102982>
- Kassa, B. Y., & Worku, E. K. (2025). The impact of artificial intelligence on organizational performance: The mediating role of employee productivity. *Journal of Open Innovation: Technology, Market, and Complexity*, 11(1), 100474. <https://doi.org/10.1016/j.joitmc.2025.100474>

- Murire, O. T. (2024). Artificial intelligence and its role in shaping organizational work practices and culture. *Administrative Sciences*, 14(12), 316. <https://doi.org/10.3390/admsci14120316>
- Peng, S., Kalliamvakou, E., Cihon, P., & Demirer, M. (2023). The impact of ai on developer productivity: Evidence from github copilot. *arXiv*. <https://doi.org/10.48550/ARXIV.2302.06590>
- Romeo, E., & Lacko, J. (2025). Adoption and integration of ai in organizations: A systematic review of challenges and drivers towards future directions of research. *Kybernetes*, 55(3), 1286–1307. <https://doi.org/10.1108/k-07-2024-2002>

Appendix A

Ethical Clearance Certificate

**CBE RESEARCH ETHICS COMMITTEE**

Dear Researchers

ETHICAL APPROVAL GRANTED FOR RESEARCH PROJECT

Decision: Clearance granted

This letter serves to confirm that the proposed research project indicated in the table below, has been reviewed by the Department of Applied Information Systems at the University of Johannesburg. Ethical clearance is hereby granted and is valid for three years, from 01 January 1900 until 31 December 2999.

Applicant	
Supervisor	
Student/staff number	
Title	
Decision date at meeting	
Reviewers	
Ethical clearance code	
Rating of application	

CODE 01 – Approved

CODE 02 – Approved with suggestions/requirements
with no re-submission

CODE 03 – Referred back

CODE 04 – Disapproved, cannot re-submit

The researcher/s may now commence with the study providing that:

1. The researcher/s will ensure that the project adheres to ethical research requirements
2. The researcher/s will be conducting the study as set out in the approval application
3. The researcher/s will ensure that project adheres to all applicable legislation, scopes of practice, professional codes of conduct and scientific standards as it pertains to the field or study.
4. The researcher/s will bring under the attention of the research ethics committee any proposed changes, concerns that arise, and unexpected ethical management issues
5. Any changes that can affect the study-related risks for participants or researchers must be reported to the committee in writing
6. Only data collected during the three valid period is permissible and valid for the study

7. No data collect before or after the three valid period is ethically valid and the Research Ethics Committee takes no responsibility for any issues arising in this situation
8. No fieldwork activities may continue after the ethical clearance has expired. A request for an extension of ethical clearance can be made in writing to the REC
9. All data collection must be done within the Government regulations of the lockdown
10. The researcher/s must be aware that COVID-19 may influence the responses and needs to be controlled for

Furthermore, the researcher/s must please address the following comments from the ethics committee:

- N/A

Prof M Name

AIS REC

aisrec@uj.ac.za

Appendix B

Appendix Chapter

B.1 Appendix section

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.

Appendix C

Another Appendix Chapter

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam lobortis facilisis sem. Nullam nec mi et neque pharetra sollicitudin. Praesent imperdiet mi nec ante. Donec ullamcorper, felis non sodales commodo, lectus velit ultrices augue, a dignissim nibh lectus placerat pede. Vivamus nunc nunc, molestie ut, ultricies vel, semper in, velit. Ut porttitor. Praesent in sapien. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Duis fringilla tristique neque. Sed interdum libero ut metus. Pellentesque placerat. Nam rutrum augue a leo. Morbi sed elit sit amet ante lobortis sollicitudin. Praesent blandit blandit mauris. Praesent lectus tellus, aliquet aliquam, luctus a, egestas a, turpis. Mauris lacinia lorem sit amet ipsum. Nunc quis urna dictum turpis accumsan semper.